LLM Backend Research for a Link-Enrichment Pipeline

Prepared October 6, 2026. Workload: ~1,500 input and ~150 output tokens per saved link; output is strict JSON (title, 2-3 sentence summary, intent bucket, 4-6 tags). Prices are USD per 1M tokens, taken from vendor pricing pages where possible. Some figures are from third-party trackers and are flagged. Verify before committing.

Summary

- Cheapest by list price: Groq gpt-oss-20b, Ministral 3 3B, then GPT-4o-mini and DeepSeek V4.1 Flash (off-peak), which cost about the same.
- Claude Haiku is more expensive than GPT-4o-mini: about 5.3x on input and 6.7x on output (3.5 Haiku); 6.7x and 8.3x for Haiku 4.5.
- Google AI Pro / Google One gives no Gemini API credits. API billing is separate.
- HTTP 402 on the Gemini API means the prepaid balance is at $0 or the billing account has a problem.
- Use a durable queue, a multi-vendor fallback chain, and circuit breakers for 402/429.

1. Pricing per 1M tokens

Model	Input	Cached input	Output	Notes
OpenAI GPT-4o-mini	$0.15	$0.075	$0.60	On model page; missing from OpenAI new pricing page
OpenAI GPT-6 Luna	$0.10	$0.01	$0.50	Current efficient OpenAI model; cheaper than 4o-mini
Claude 3.5 Haiku	$0.80	$0.08	$4.00	Batch $0.40 / $2.00
Claude 3 Haiku	n/a	n/a	n/a	Retired April 20, 2026
Claude Haiku 4.5	$1.00	$0.10	$5.00	Current Haiku
Gemini 3.5 Flash-Lite	$0.30	$0.03	$2.50	Batch/Flex $0.15 / $1.25
Gemini 3.1 Flash-Lite	$0.25	$0.025	$1.50	Batch/Flex $0.125 / $0.75
Gemini 3 Flash Preview	$0.50	$0.05	$3.00	
Gemini 3.8 Flash	$0.75	$0.075	$3.75	Promo through Dec 31, 2026; then $1.50 / $7.50
Ministral 3 3B	$0.10	n/a	$0.10	Third-party source
Ministral 3 8B	$0.15	n/a	$0.15	Third-party source
Mistral NeMo	$0.15	n/a	$0.15	Third-party source
DeepSeek V4.1 Flash (off-peak)	$0.15	$0.003	$0.60	Peak rates are 2x
Groq gpt-oss-20b	$0.075	n/a	$0.30	
Groq Llama 3.1 8B	$0.05	n/a	$0.08	Reported enterprise-only since Aug 16, 2026

- Gemini 2.0 Flash is not on Google current pricing page. Gemini output pricing includes thinking tokens; set thinking to minimal/off for this task.
- DeepSeek peak hours: 01:00-04:00 and 06:00-10:00 UTC on weekdays. Older $0.14/$0.28 figures are outdated.
- Free options: OpenRouter :free models allow 20 RPM and 50 requests/day (1,000/day after a one-time $10 top-up). Groq free tier: 30 RPM, 1,000 RPD, 8,000 TPM, 200,000 tokens/day (about 120 links/day at ~1,650 tokens each).

2. Claude Haiku vs GPT-4o-mini

Comparison	Input multiplier	Output multiplier
Claude 3.5 Haiku vs GPT-4o-mini	0.80 / 0.15 = 5.3x	4.00 / 0.60 = 6.7x
Claude Haiku 4.5 vs GPT-4o-mini	1.00 / 0.15 = 6.7x	5.00 / 0.60 = 8.3x

For your workload, 3.5 Haiku costs about 5.7x as much as GPT-4o-mini overall. Claude 3 Haiku is retired.

3. Cost for your workload (1,500 in / 150 out, uncached, standard rates)

Model	Per link	1,000 links	10,000 links
Groq gpt-oss-20b	$0.000158	$0.16	$1.58
Ministral 3 3B	$0.000165	$0.17	$1.65
Ministral 3 8B	$0.000248	$0.25	$2.48
GPT-4o-mini	$0.000315	$0.32	$3.15
DeepSeek V4.1 Flash (off-peak)	$0.000315	$0.32	$3.15
DeepSeek V4.1 Flash (peak)	$0.000630	$0.63	$6.30
Gemini 3.1 Flash-Lite	$0.000600	$0.60	$6.00
Gemini 3.5 Flash-Lite	$0.000825	$0.83	$8.25
Claude 3.5 Haiku	$0.001800	$1.80	$18.00
Claude Haiku 4.5	$0.002250	$2.25	$22.50

- Gemini Batch/Flex halves cost (3.5 Flash-Lite about $4.13 per 10,000 links).
- Thinking tokens on Gemini can inflate output cost well beyond 150 tokens.
- Caching helps little because most of the input is page content that varies.
- At 10,000 links the gap between cheapest and Claude is about $16, so quality and reliability matter more than price at this scale.

4. Google ecosystem and API gotchas

Google AI Pro / Gemini Advanced and the API. No API credits. Consumer subscriptions and AI Studio/Cloud API billing are separate, and a separate billing account is required. Google AI Pro lists $10/month in Google Cloud credits through the Google Developer Program, but Google billing docs say only some Cloud credits apply to the Gemini API and the Welcome credit does not. Not confirmed for the $10 credit; check in the Cloud console.

Causes of HTTP 402 (prepay credits depleted).

- Prepay balance reached $0: all keys in all projects on that billing account stop until credits are added.
- Delinquent balance, declined payment or expired card on the same Cloud billing account, even for unrelated services.
- Abandoned Prepay setup flow leaving the account unchargeable.
- Billing lag of about 10 minutes can push balances negative before cutoff.
- Hitting the tier monthly cap (Tier 1 is $250) pauses service until the 1st of the month.
- Prepaid credits expire after 12 months; minimum purchase $5; auto-reload with a monthly limit is available.

Free-tier rate limits. Google no longer publishes a fixed public table; limits are shown per project and model in AI Studio, are not guaranteed, are per project (not per key), and reset at midnight Pacific. A third-party April 2026

snapshot lists Gemini 3.1 Flash-Lite at 15 RPM / 1,000 RPD and Gemini 3 Flash at 10 RPM / 1,500 RPD (approximate). Free-tier content may be used to improve Google products, so do not send real user data through it in production.

5. Production architecture recommendation

- Durable queue (Redis/BullMQ, SQS, or Postgres jobs). Save the link immediately with a placeholder title (og:title) and status pending_enrichment; never block the UI on the LLM.
- Provider chain behind one interface: primary GPT-4o-mini or Gemini 3.1 Flash-Lite; secondary a different vendor (DeepSeek V4.1 Flash or Ministral); tertiary OpenRouter with funded balance; last resort Groq gpt-oss-20b or a rule-based fallback.
- 429: exponential backoff with jitter, honor Retry-After, per-provider token bucket. 402/401: do not retry; open circuit breaker, fail over, alert. 5xx/timeouts: retry twice, then fail over.
- Credit protection: auto-reload with a monthly cap, budget limits, balance alerts, and a dedicated Cloud billing account for Gemini so unrelated declined payments cannot stop it.
- Quality/cost: JSON-schema structured output with validation and one repair retry; cache by canonical URL hash; truncate input to the token budget; send backfills via Batch APIs; log provider/model per row.
- Before committing, run ~200 real links through your top two candidates and compare JSON validity and tag quality.

Sources

Google Gemini API pricing, billing and rate-limit docs (ai.google.dev); OpenAI API pricing and GPT-4o-mini model page; Anthropic Claude pricing and model-deprecation docs; DeepSeek API pricing docs; Mistral pricing via Puter (June 2026); Groq docs and third-party Groq trackers; OpenRouter limits docs and free-API comparison; Google One AI plans page and Google AI developer forum / Stack Overflow threads on AI Pro vs API billing; PE Collective Gemini free-tier guide.